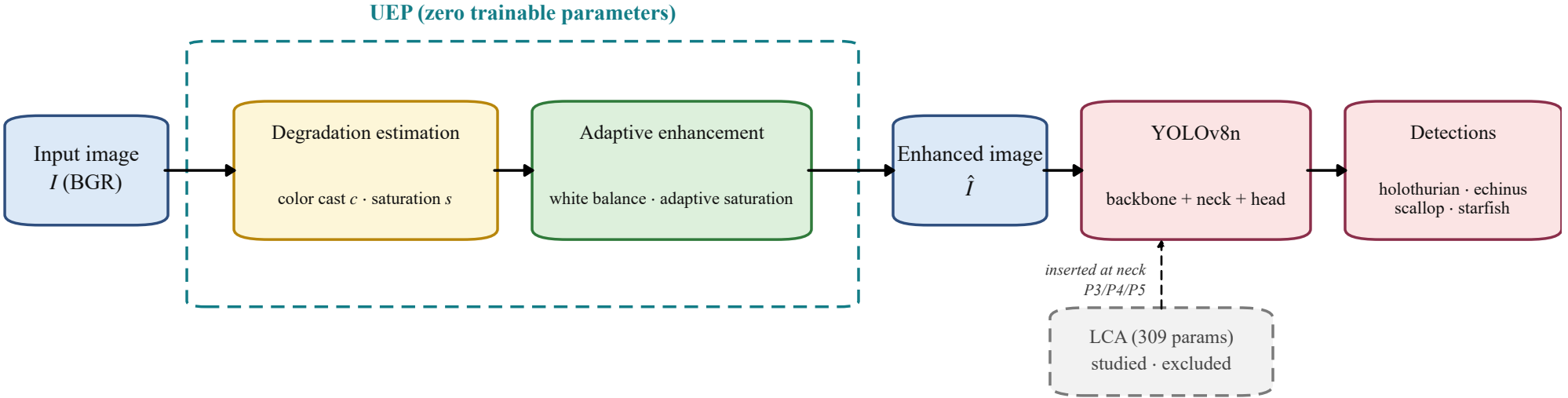

UEP (zero trainable parameters)
Input image
I (BGR)
Degradation estimation
color cast c · saturation s
Adaptive enhancement
white balance · adaptive saturation
Enhanced image
Î
YOLOv8n
backbone + neck + head
Detections
holothurian · echinus
scallop · starfish
inserted at neck
P3/P4/P5
LCA (309 params)
studied · excluded